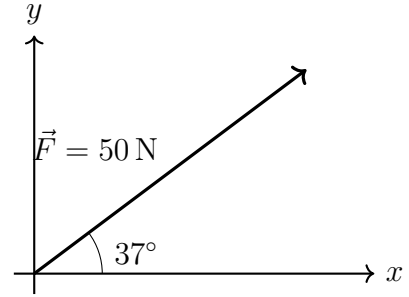


Problem Set: Vectors, Trigonometry, and Frames of Reference

1. A force $\vec{F}$ of magnitude 50 N is directed at 37° above the horizontal, as shown.

(a) Find the horizontal component F_x .

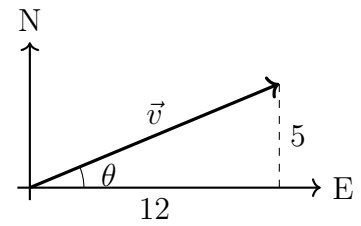
(b) Find the vertical component F_y .



2. A car's velocity has components $v_x = 12 \text{ m/s}$ (east) and $v_y = 5 \text{ m/s}$ (north).

(a) Find the speed $|\vec{v}|$.

(b) Find the direction θ , measured north of east, to the *nearest degree*.



3. **Frame of reference — on a train.** A train travels east at 25 m/s relative to the ground. A passenger walks toward the front of the train at 1.5 m/s relative to the train.

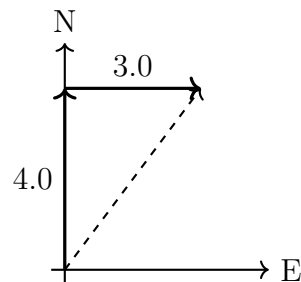
(a) What is the passenger's velocity relative to the ground?

(b) A car drives *west* at 15 m/s relative to the ground. What is the passenger's velocity relative to the car?

4. **Frame of reference — crossing a river.** A boat heads due north across a river, moving at 4.0 m/s relative to the water. The river itself flows east at 3.0 m/s relative to the ground.

(a) Find the boat's speed relative to the ground.

(b) Find the angle of the boat's motion, measured east of north, to the *nearest degree*.



5. **Energy unit conversions.** A typical chocolate bar contains about 1.0×10^6 J of food energy.

(a) Convert this to kilocalories. (1 kcal = 4184 J.)

(b) Convert this to kilowatt-hours. (1 kWh = 3.6×10^6 J.)

(c) How many electron-volts is 1.0×10^6 J? (1 eV = 1.6×10^{-19} J.)